

BASED ON THE PAPER

Grant Selection Design and Start-Up Innovation

Evidence from Design-Dependent Subsidy Effects

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Curriculum Vitae

▶ School

Sidcot Independent School England	2012–2014
International Baccalaureate	

▶ University

Technical University of Munich B.Sc. Technology and Management	2014–2020
Technical University of Munich M.Sc. Technology and Management	2020–2024

▶ Professional Experience

ColInvest Corporate Finance Analyst	2021–2024
agentix GmbH Co-Founder & COO	2024–2025
Independent Management Consultant	2024–present

Research Question

How can you improve investments into Start-Ups and SMEs through legislative change of Investment Modalities?

- ▶ AI at the forefront of Innovation
- ▶ *“Automating Cognitive Workflows”*
- ▶ Recipient of EXIST Grant

Core Question

Do the selection signals used in public SME and start-up funding programmes generate technological innovation, observable spillover, and economic value?

- ▶ **Motivation:** Private returns to innovation fall short of social returns (Arrow, 1962) — the appropriability problem is sharpest for early-stage firms
- ▶ **Gap:** Literature evaluates *whether* grants work; not *which selection signals* generate which outcomes
- ▶ **Contribution:** Shifts programme evaluation from beneficiary status to the economic content of the selection rules

Two-step design

1. **Identify the selection rule** — code programme documents, scoring rubrics and eligibility criteria; recover implied signal weights; compare stated vs. applied weights
2. **Estimate causal effects** — regression-discontinuity around score / budget cut-offs; applicants just above and below the threshold are otherwise comparable

Three outcome layers

- ▶ **Technological Innovation** — patents, product/process launches, technology-class novelty
- ▶ **Observable Spillover** — licensing, research collaborations, technology diffusion
- ▶ **Economic Value** — employment, enterprise survival, cluster/sector growth

Empirical Framework

- ▶ **Unit of analysis:** enterprise–application pair.
- ▶ **Object of interest:** observable application signals used in the funding decision.
- ▶ **Treatment margin:** applicants just above and just below score or budget cut-offs.
- ▶ **Preferred design:** regression discontinuity around selection thresholds.
- ▶ **Outcome hierarchy:**
 - ▷ technological innovation
 - ▷ observable spillover
 - ▷ local economic value

Research Proposal: Data and Methodology

Data Sources

- ▶ Programme documents, scoring rubrics, beneficiary lists (EU transparency mandate)
- ▶ CORDA / ORBIS — applicant-level award data
- ▶ EPO / Patstat — patent records
- ▶ Business registers & regional employment data
- ▶ EIC Accelerator score-level data

Identification Strategy

- ▶ **Preferred:** Regression discontinuity at score / budget cut-offs (Imbens & Lemieux, 2008)
- ▶ **Alternative:** DiD / event study across calls, regions, countries (Callaway & Sant'Anna, 2021)
- ▶ **Threats:** Self-selection into application, score manipulation near cut-offs, survivorship bias

Expected Contribution

- ▶ **Empirical:** estimate whether selection signals generate the outcomes used to justify public support.
- ▶ **Conceptual:** separate the effect of receiving a grant from the design of the rules that allocate grants.
- ▶ **Policy-relevant:** identify where programme objectives and actual selection practice diverge.
- ▶ **PhD logic:** the project can begin as one empirical paper on grant selection design and develop into a broader research agenda on innovation policy.

Backup: Data Sources

- ▶ Programme documents, eligibility criteria, scoring rubrics and beneficiary lists.
- ▶ Applicant-level award data from CORDA / ORBIS where accessible.
- ▶ Patent data from EPO / PATSTAT.
- ▶ Firm accounts, business registers and regional employment data.
- ▶ Score-level data from programmes such as the EIC Accelerator where available.

Backup: Identification Risks

- ▶ Self-selection into application.
- ▶ Score manipulation near cut-offs.
- ▶ Missing or incomplete score-level data.
- ▶ Survivorship bias in firm-level outcomes.
- ▶ Outcome correlation across innovation, spillover and economic-value layers.